



“Silver paste represents 10-15% of photovoltaic cell manufacturing costs, and sustained increases can put pressure on margins and near-term project economics across the value chain”

GAGAN CHANANA
JOINT MD & CEO, JAKSON
SOLAR MODULES AND CELLS



“Manufacturers are increasingly offsetting part of [the] impact through efficiency improvements and material optimisation rather than passing costs directly to customers”

VINAY THADANI
DIRECTOR & CEO, GREW SOLAR

from January 1, 2026, post the ban on export of critical minerals earlier, has further increased the price of the white metal.

Aditya Pyasi, Chief Executive Officer, Indian Wind Turbine Manufacturers Association (IWTMA), says India's copper refining capability has weakened over the past decade, primarily after the 2018 closure of the 400,000 tonne per year Tuticorin smelter, which alone accounted for about 35-40% of national refined copper capacity and had previously helped make the country a net exporter of copper cathodes.

“Its high time we take decisive steps for self-reliance in raw materials,” he says.

Companies, meanwhile, have cut down on use of silver as they ramp up solar capacity to meet the growing domestic demand. Emmvee, a solar energy company, said during the Q3FY26 investor call that silver paste now accounts for a far smaller share of total cell costs than in previous years, following sustained reductions in consumption through process innovation and R&D.

“Silver consumption has already come down by more than 50-60% over the last three to four months, and we are targeting another 40% reduction going forward,” the management said, adding that silver paste currently costs about two cents per watt or roughly 13% of the cell price.

Technological advances such as silver-coated copper powders and zero-busbar designs are reducing the amount of silver required per solar cell but these efficiency gains are being offset by scale. Adoption of highly-efficient solar cells, such as Tunnel Oxide Passivated Contact and Heterojunction technologies, is also acting as a counter-force by increasing the demand for silver.

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As metallic headwinds for the solar industry tighten margins in the near term, manufacturers are increasingly offsetting a part of this impact through efficiency improvements and material optimisation rather than passing costs directly to customers.

Prashant Mathur, CEO, Saatvik Green Energy Ltd, says they are mitigating the volatility through deeper vertical integration, tighter procurement controls, and technology optimisation.

India is also marching ahead on solar installations. Besides utility scale projects, the rooftop solar scheme is targeting four million households by March 2026, and 10 million by 2027, to make clean energy a part of everyday life while reducing the dependence on conventional power. The scheme links solar adoption to household savings and offers graded rooftop-solar subsidies and easy, collateral-free loans facilitated through a national digital portal.

Chanana of Jackson Solar says ongoing advances in cell design and metallisation techniques, including finer line